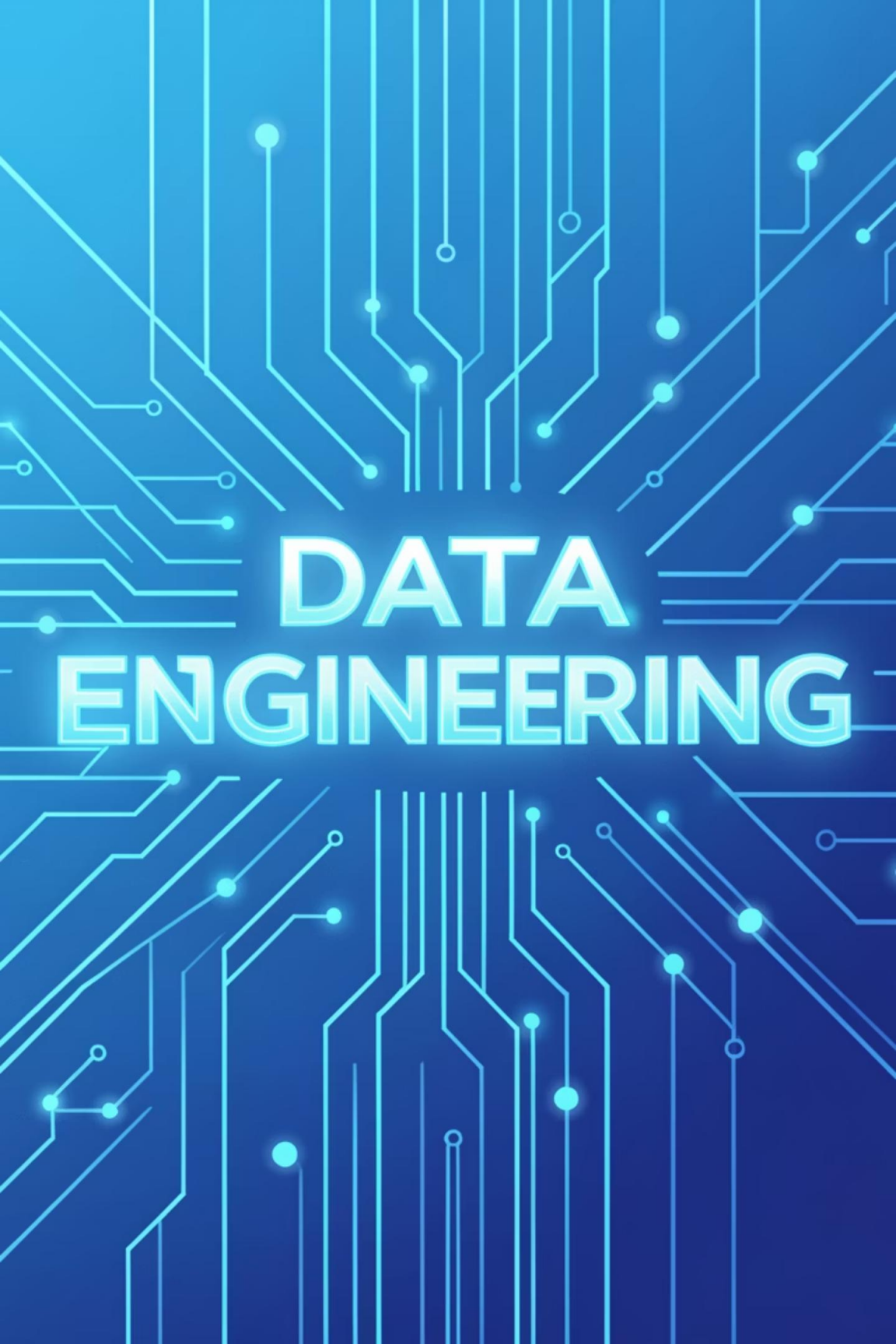




Data Engineering

Spring '26 – DS471

January 22, 2026

Lecture 2

Logistics: Your Course Essentials

Class Timings

Monday, Tuesday: 11:30 AM – 12:20 PM, LH6

Friday: 11:00 AM – 11:50 AM, LH6

Office Hours

Monday, Tuesday: 10:00 AM – 11:00 AM, S18 2nd Floor NAB or by appointment.

Credit Hours

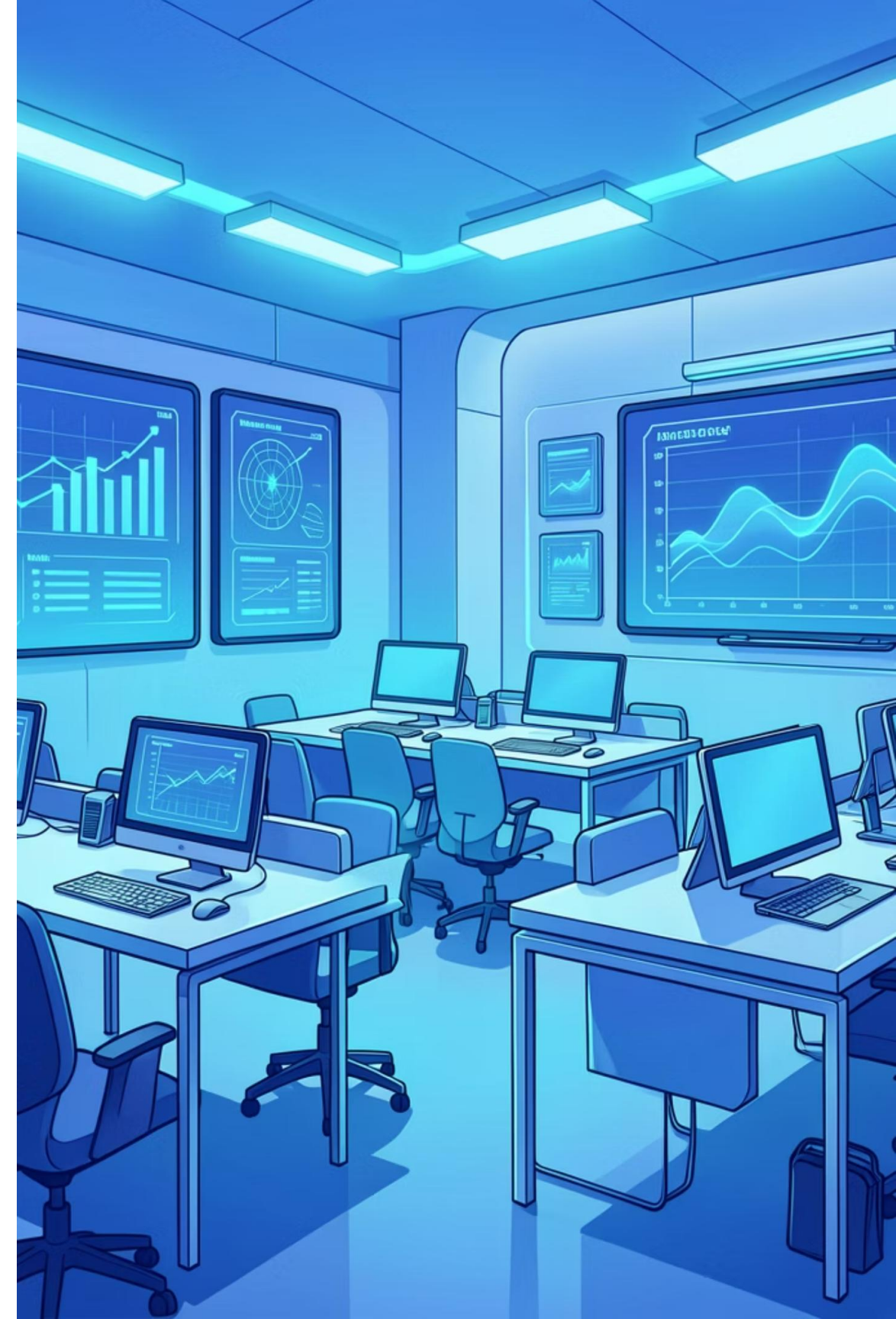
2 hours (Lectures) + 1 hour (Practical Session)

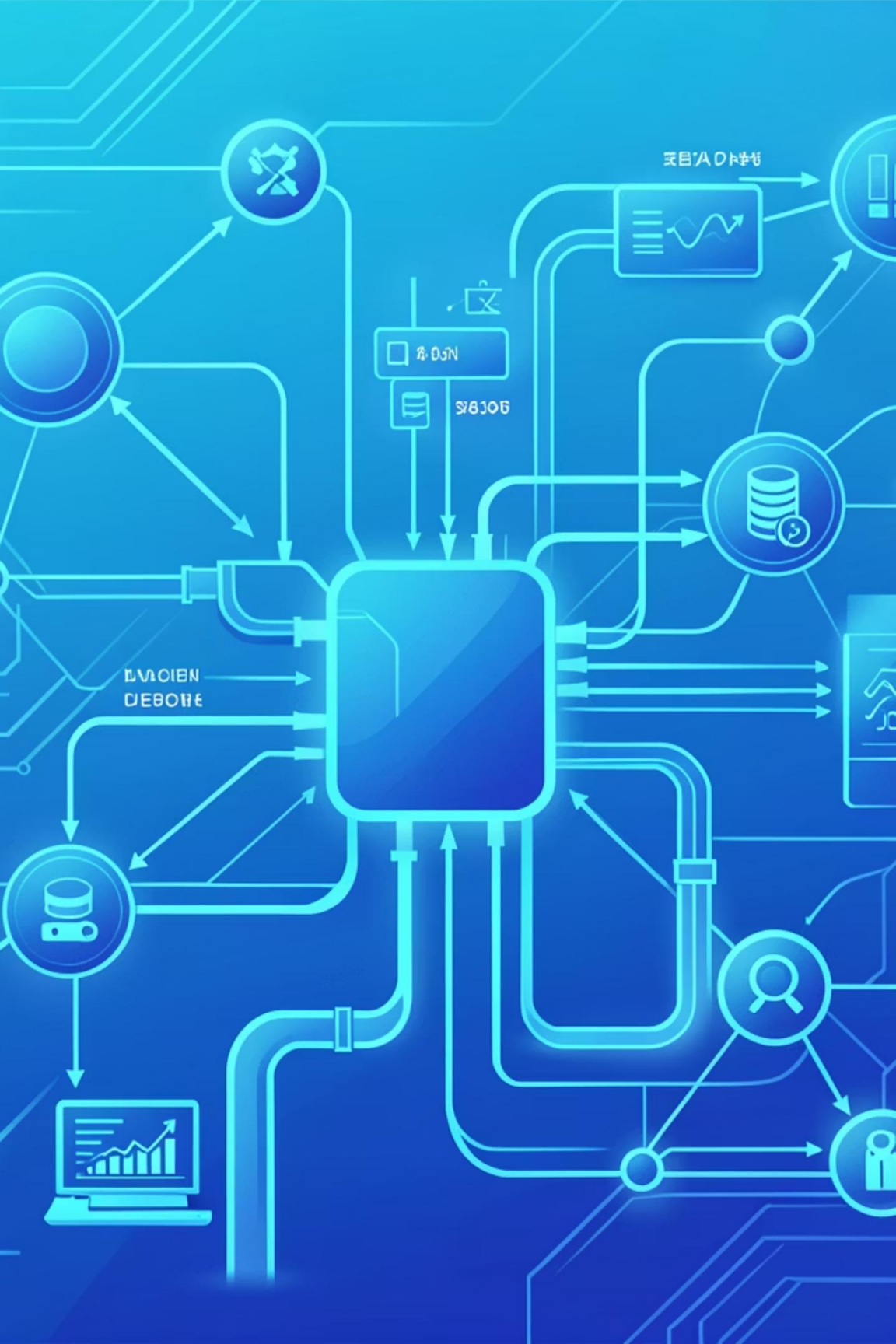
Contact & Support

MS Teams: [TBD](#)

Email: farah.saeed@giki.edu.pk

WhatsApp: Reach out for quick queries





Understanding Data Engineering & The Data Engineer

Data Engineering

The discipline of developing, implementing, and maintaining robust systems and processes. It transforms raw data into high-quality, consistent information to support critical downstream uses like analysis and machine learning.

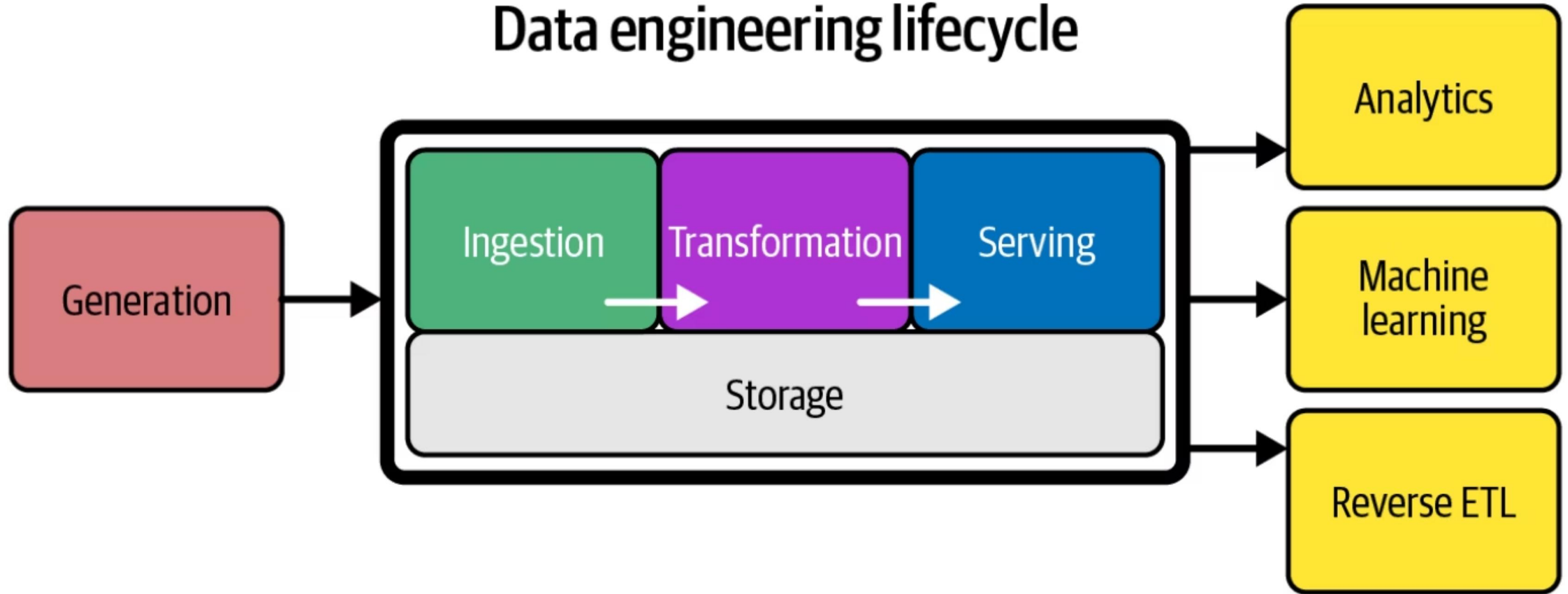
Data Engineer

A specialist who manages the entire data engineering lifecycle. This role spans from acquiring data from diverse source systems to delivering processed data for various applications, including advanced analytics and machine learning models.

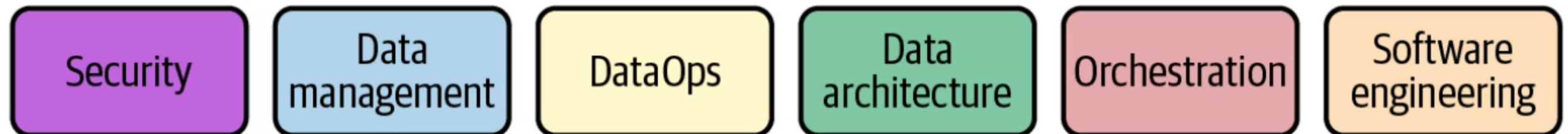
The Data Engineering Lifecycle in Action

The data engineering lifecycle illustrates the continuous journey of data from its origin to its final consumption. Each stage is crucial for ensuring data quality, accessibility, and utility.

Data engineering lifecycle



Undercurrents:



Key Stages of the Data Engineering Lifecycle

01

Generation

Data is created from various sources, such as applications, IoT devices, user interactions, and external systems.

02

Storage

Raw and processed data are stored in appropriate data lakes, warehouses, or databases, optimized for retrieval and security.

03

Ingestion

Data is collected and imported from diverse sources into the storage systems, often involving real-time or batch processes.

04

Transformation

Raw data is cleaned, structured, and enriched to meet the requirements of analytical models and applications.

05

Serving

Processed data is made accessible to end-users, business intelligence tools, and machine learning models for insights and decision-making.

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Evolution of the Data Engineer Role

- 1 — 1980-2000: Data Warehousing to the Web
Emergence of business data warehouses, IBM's relational databases (SQL), and scalable analytics with MPP databases. The 1990s saw expensive, monolithic infrastructure.
- 2 — Early 2000s: Birth of Contemporary Data Engineering
The "big data" era begins with Google File System and MapReduce, followed by Yahoo's Apache Hadoop, laying the groundwork for distributed processing.
- 3 — 2010s: Cloud Computing & Scalability
The rise of cloud services like Amazon EC2, S3, and DynamoDB revolutionized data storage and processing, offering infinitely scalable solutions.
- 4 — 2020s: Modern Data Stack
Current trends focus on the modern data stack, a collection of open-source and third-party tools designed to streamline data analysis and engineering workflows.